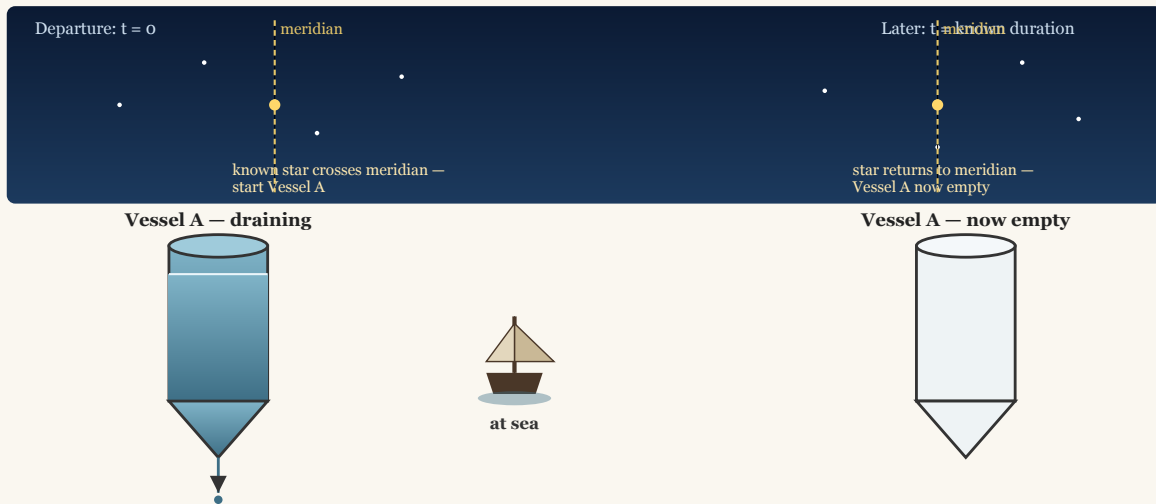


Van Langren's Water-Clock Method for Longitude (1634)

Two identical copper vessels, timed against a star's return to the meridian



1. At departure, note a known star crossing the meridian — start Vessel A draining.
2. From the stars, the exact time it should take Vessel A to empty is already known.
3. If the star returns to the meridian later or earlier than the vessel empties, the mismatch reveals how far east or west the ship has traveled — its longitude.

Diagram reconstructed from published translations — not a facsimile of Van Langren's coded original (1634).